

IEEE 802.11 WiFi

Network Simulation

Complete Technical Walkthrough

Stage 1 & Stage 2 — Combined

Protocol	IEEE 802.11 b/g/n (WiFi) at 2.4 GHz / 5 GHz
Modulations	BPSK · QPSK · 16-QAM · 64-QAM
Channel Model	AWGN + Indoor Log-Distance + Rayleigh Fading
Key Metrics	BER · Throughput · SNR · Path Loss · Data Rate
Simulation Tool	Python (NumPy, SciPy, Matplotlib, NetworkX)
Network Size	Stage 1: 3 nodes (AP+2 STA) · Stage 2: 15 nodes (1 AP+14 STA)
Scenarios	Stage 1: Normal / Relay Fail / Interference · Stage 2: Normal / War-like
Institution	Wireless Networks (WN PBL) — VIIT Pune

STAGE 1 — Basic WiFi Simulation

STAGE 2 — Advanced Simulation

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STAGE 1

Basic WiFi Simulation

Point-to-Point Link & 3-Node Relay Network

1 Simulation Architecture & Setup

The simulation is divided into two parts. **Part 1** models a single point-to-point WiFi link between an Access Point (AP) and a Station (STA) 50 metres apart indoors. **Part 2** extends this to a 3-node relay network.

Network Architectures

Part	Topology	Description
Part 1	AP — 50 m — STA	Single point-to-point indoor link
Part 2	AP — 25 m — STA-1 — 25 m — STA-2	3-node relay: AP transmits via relay to destination

Core Parameters

Parameter	Value	Technical Meaning
Carrier Frequency	2.4 GHz	Standard WiFi band
TX Power	20 dBm	100 mW — typical AP output
Distance (Part 1)	50 m	Indoor link distance
Noise Floor	-77 dBm	Receiver sensitivity baseline
Path Loss Exp. n	3	Indoor office — walls & furniture
Bits Simulated	10,000 bits	For accurate BER estimation
Random Seed	42	Reproducible results

Libraries Used

Library	Role
numpy	Numerical arrays, random bit generation
scipy.special.erfc	Complementary error function for BER formulas
matplotlib	Result visualisations and plots

2 Indoor Path Loss Model

In a real indoor WiFi environment, signals are weakened not only by distance but also by walls, furniture, and reflections. The simulation uses the **Log-Distance Path Loss Model** — far more realistic than free-space for indoor WiFi.

$$PL(d) = PL_0 + 10 \times n \times \log_{10}(d)$$

where $PL_0 = 40 \text{ dB}$ (reference loss at 1 m at 2.4 GHz), d is distance in metres, and n is the path loss exponent.

Environment	Path Loss Exponent (n)	Typical Use
Free Space	2	Open air, line-of-sight
Indoor Office	3 (used here)	Offices, homes, walls
Dense Indoor	4-6	Factories, hospitals

◆ Key result at 50 m

Indoor path loss = 82.1 dB vs free-space = 60 dB. The indoor environment adds about 22 dB of extra attenuation — roughly 150x more power lost indoors.

3 Received Power & SNR Calculation

Once path loss is known, received power and Signal-to-Noise Ratio follow directly by subtraction on the dB scale.

$$\text{RX Power (dBm)} = \text{TX Power} - \text{Path Loss_indoor} \quad \text{SNR (dB)} = \text{RX Power} - \text{Noise Floor}$$

Parameter	Value
TX Power	20 dBm
Indoor Path Loss	82.1 dB (at 50 m, n=3)
RX Power	20 - 82.1 = -62.1 dBm
Noise Floor	-77 dBm
SNR (dB)	-62.1 - (-77) = 14.9 dB
SNR (linear)	$10^{(14.9/10)} = 30.9$

◆ Why SNR matters

An SNR of ~15 dB at 50 m is solid for lower modulations (BPSK/QPSK) but marginal for 64-QAM, which demands ~25 dB for reliable operation. This is precisely why WiFi rate adaptation is essential.

4 BER Formulas by Modulation

Each modulation scheme packs a different number of bits per symbol. More bits per symbol means higher data rate but greater noise sensitivity, resulting in higher BER for the same SNR. All formulas use the Complementary Error Function $\text{erfc}()$.

Modulation	Bits/Symbol	BER Formula	SNR needed for BER < 1%
BPSK	1	$0.5 \times \text{erfc}(\sqrt{\text{SNR}})$	~4 dB
QPSK	2	$0.5 \times \text{erfc}(\sqrt{\text{SNR}})$	~4 dB
16-QAM	4	$(3/8) \times \text{erfc}(\sqrt{2 \times \text{SNR}/5})$	~11 dB
64-QAM	6	$(7/24) \times \text{erfc}(\sqrt{\text{SNR}/7})$	~17 dB

◆ Why BPSK and QPSK share the same BER formula

QPSK carries 2 bits/symbol but encodes them on two orthogonal (90 deg perpendicular) signal dimensions (I and Q). Each dimension is independently BPSK, so each bit faces the same noise margin — identical BER per bit, but double the speed.

5 IEEE 802.11 Data Rate Analysis

The simulation tests all 8 standard 802.11 modes at the computed SNR (~15 dB at 50 m) and measures BER and effective throughput for each.

Standard	Modulation	Data Rate	Typical BER	Throughput	Status
802.11b	BPSK	1 Mbps	~0.000000	~1.0 Mbps	EXCELLENT
802.11b	QPSK	2 Mbps	~0.000000	~2.0 Mbps	EXCELLENT
802.11g	BPSK	6 Mbps	~0.000000	~6.0 Mbps	EXCELLENT
802.11g	QPSK	12 Mbps	~0.000000	~12.0 Mbps	EXCELLENT
802.11g	16-QAM	24 Mbps	~0.000100	~24.0 Mbps	GOOD
802.11g	16-QAM	36 Mbps	~0.001200	~35.9 Mbps	GOOD
802.11n	64-QAM	48 Mbps	~0.015000	~47.3 Mbps	MARGINAL
802.11n	64-QAM	54 Mbps	~0.015000	~53.2 Mbps	MARGINAL

Note: BER values vary slightly per run due to random noise simulation. Values shown are representative at SNR ≈ 15 dB.

6 Adaptive Rate Selection

Real WiFi devices continuously measure SNR and select the highest data rate that keeps BER below 1%. The simulation replicates this by scanning all modes and choosing the highest throughput among those with BER < 1%.

SNR Range	Selected Mode	Reason
> 17 dB	802.11g 16-QAM 36 Mbps	16-QAM BER well below 1% — optimal
11–17 dB	802.11g 16-QAM 24 Mbps	64-QAM BER exceeds 1%; 16-QAM works
4–11 dB	802.11b QPSK 2 Mbps	16-QAM too noisy; BPSK/QPSK viable
< 4 dB	No Link	No modulation survives

◆ At 50 m indoor (SNR ≈ 15 dB)

WiFi adapts to 802.11g 16-QAM at 36 Mbps as the optimal trade-off between speed and reliability. 64-QAM modes (48/54 Mbps) are rejected because their BER exceeds 1%.

7 3-Node Relay Network

Part 2 extends the simulation to a 3-node topology: an Access Point (AP), a Relay station (STA-1), and a Destination (STA-2). STA-1 sits halfway at 25 m from both AP and STA-2.

Link	Distance	Path Loss	RX Power	SNR
AP -> STA-1	25 m	~73.5 dB	-53.5 dBm	~23.5 dB
STA-1 -> STA-2	25 m	~73.5 dB	-53.5 dBm	~23.5 dB
AP -> STA-2 (direct)	50 m	~82.1 dB	-62.1 dBm	~14.9 dB

◆ Why relay helps

The relay provides 8.6 dB better SNR per hop versus the direct 50 m link. In log-distance models, halving distance removes 10 x n dB of loss — this directly enables higher data rates and lower BER.

8 Three Fault Scenarios

Scenario	Setup	SNR/hop	BER (e2e)	Throughput	Quality
1 — Normal Relay	AP->STA-1->STA-2 (2x25m, 16-QAM)	~23.5 dB	~0.000010	~24 Mbps	EXCELLENT
2 — Relay Failed	AP->STA-2 direct (50m), rate adapt	~15 dB	~0.001000	~12-24 Mbps	GOOD
3 — Interference	+10 dB noise (microwave/Bluetooth)	~13.5 dB	~0.005000	~20-22 Mbps	USABLE

Scenario Details

◆ Scenario 1 — Normal Relay

Both hops at 25 m achieve SNR ~23.5 dB. BER is very low and throughput is close to nominal 24 Mbps. This is the baseline best-case for the relay path.

◆ Scenario 2 — Relay Failed

STA-1 is down; signal must travel the full 50 m. SNR drops to ~15 dB. WiFi performs rate adaptation and auto-selects a lower data rate. The link survives but at reduced throughput.

◆ Scenario 3 — High Interference

A nearby microwave oven or Bluetooth device raises the noise floor by +10 dB to -67 dBm. All link SNRs drop by 10 dB simultaneously, pushing BER up and forcing lower data rates.

9 Key Findings — Stage 1

◆ Higher data rate = higher BER

64-QAM (54 Mbps) requires ~25 dB SNR for reliable operation. At 15 dB it degrades badly. BPSK works fine down to ~4 dB. This is the fundamental modulation trade-off.

◆ Indoor path loss far exceeds free-space

At 50 m, free-space loss = 60 dB but indoor = 82 dB. Walls and obstacles add ~22 dB extra — about 150x more power lost indoors.

◆ Relay nodes dramatically improve SNR

Splitting 50 m into two 25 m hops improves SNR by 8.6 dB per hop. In log-distance models, halving distance removes $10 \times n$ dB of path loss.

◆ WiFi rate adaptation is essential

802.11 continuously selects the best modulation for current conditions. When relay fails, it drops to a lower rate rather than losing the link entirely.

◆ Interference compounds quickly

+10 dB noise (microwave ovens, Bluetooth) reduces SNR by 10 dB on every link. This can push marginal links from GOOD to POOR instantly.

◆ BER 1% is the practical threshold

WiFi refuses to use a data rate when $BER > 1\%$. This appears in rate adaptation throughout.

◆ Throughput != nominal data rate

Effective throughput = $\text{nominal_rate} \times (1 - BER)$. A "54 Mbps" link at 5% BER only delivers ~51 Mbps.

STAGE 2

Advanced Simulation

15-Node Random Topology + Rayleigh Fading + War-like Scenarios

10 Libraries & Setup

Stage 2 uses four Python libraries. NetworkX is added for graph/topology support.

Library	Purpose
numpy (np)	Numerical arrays, random coordinates, Euclidean distance, logarithms
matplotlib.pyplot (plt)	Topology maps and comparative performance charts
scipy.special.erfc	Complementary Error Function for theoretical BER in AWGN channels
networkx (nx)	Graph/topology generation support for multi-node networks

◆ Random Seed

`np.random.seed(42)` fixes the RNG so Rayleigh fading events and STA coordinates are identical on every run — making results fully reproducible and comparable.

11 Simulation Parameters

All physical constants are based on real-world IEEE 802.11 specifications. Stage 2 uses 50,000 bits per node for higher BER statistical accuracy.

Parameter	Value	Technical Meaning
num_bits	50,000	Bits per node per scenario — more bits = more accurate BER
carrier_freq	2.4 x 10 ⁹ Hz	WiFi 2.4 GHz band. Wavelength = 0.125 m
tx_power_dBm	20 dBm	Router TX power = 100 mW. $P(\text{dBm}) = 10 \times \log_{10}(P_{\text{mW}})$
max_distance	50 m	Simulation area radius
noise_power_dBm	-77 dBm	Baseline receiver noise floor (thermal noise)
PL_0	40 dB	Reference path loss at 1 m from AP at 2.4 GHz
n_exponent_normal	3	Indoor path loss exponent for Scenario A

Key dB Conversions

Expression	Formula	Example
Power -> dBm	$P(\text{dBm}) = 10 \times \log_{10}(P_{\text{mW}})$	100 mW = 20 dBm
dBm -> Power	$P(\text{mW}) = 10^{(P_{\text{dBm}}/10)}$	20 dBm = 100 mW
Ratio -> dB	$P(\text{dB}) = 10 \times \log_{10}(\text{ratio})$	150x = 21.8 dB
dB -> Linear	$\text{snr}_{\text{lin}} = 10^{(\text{snr}_{\text{dB}}/10)}$	15 dB = 31.6 (linear)

12 Dynamic Network Topology Generation

The simulation generates a random 15-node network. The AP is fixed; the 14 STAs are scattered randomly across the 50x50 m space — modelling a realistic multi-device indoor deployment.

Node Placement

Each STA is assigned a random (x, y) coordinate inside the 50x50 m space using `np.random.uniform(10, 50)`. The AP is fixed at (5, 5) — near one corner, typical of a wall-mounted router.

```
coords = np.random.uniform(10, area_size, size=(num_nodes-1, 2))
ap_coord = np.array([[5, 5]])
all_coords = np.vstack((ap_coord, coords))
```

Distance Calculation

For every STA, the Euclidean distance from the AP is computed. This distance drives all subsequent calculations — path loss, SNR, BER, and rate selection all depend on it.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

```
np.linalg.norm(coord - pos['AP'])
```

◆ Example

AP at (5, 5), STA at (45, 45) -> $d = \sqrt{40^2 + 40^2} = 56.6$ m. This node is outside the 50 m radius and will likely show No Link in normal mode.

13 Indoor Path Loss Model (Extended)

Stage 2 extends the path loss model to support two frequency bands (2.4 GHz and 5 GHz) and two path loss exponents ($n=3$ normal, $n=4$ war-like), enabling frequency hopping as an anti-jamming response.

$$PL(d) = PL_0 + 10 \times n \times \log_{10}(d) \quad PL_0 = 40 \text{ dB (2.4 GHz)} \mid PL_0 = 46 \text{ dB (5 GHz)}$$

```
def get_indoor_path_loss(dist, n_exp, freq_band='2.4GHz'):
    pl_0 = 40 if freq_band == '2.4GHz' else 46
    return pl_0 + 10 * n_exp * np.log10(dist)
```

Environment	n	Path Loss at 50 m	Typical Use
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Free Space	2	74 dB	Open air, line-of-sight
Normal Indoor	3	91 dB	Scenario A — offices, homes
Dense Indoor	4	108 dB	Scenario B — rubble, concrete
Very Dense	5-6	124+ dB	Factories, underground

◆ Indoor Scenario B vs Free Space at 50 m

$108 - 74 = 34$ dB extra loss at $n=4$ — about 2500x more power lost than free space. This is why WiFi range collapses drastically in rubble-filled or dense concrete environments.

14 Rayleigh Fading — Channel Impairment

Path loss is a predictable, average effect. Rayleigh fading is the unpredictable, moment-to-moment variation on top of that average — caused by multipath propagation.

What Causes Rayleigh Fading

In any indoor environment, a radio signal does not travel in a single straight line. It bounces off walls, floors, ceilings, furniture, and people. Dozens of reflected copies arrive at the receiver from different directions at slightly different times — this is multipath propagation. These copies combine at the antenna: sometimes constructively (stronger signal), sometimes destructively — causing a deep fade of 20-30 dB in an instant, even if the device has not moved.

Property	Path Loss	Rayleigh Fading
Nature	Predictable, deterministic	Random, stochastic, instantaneous
Cause	Distance + wall absorption	Multipath reflected copies cancelling
Typical scale	Tens of dB over tens of metres	+/-5 to -30 dB in milliseconds
Applied in	Both scenarios (always)	Scenario B only (apply_fading=True)

Mathematical Model

Signal amplitude follows a Rayleigh distribution. Signal power follows an Exponential distribution. The code models this directly:

```
def get_rayleigh_fading_dB():
    power_gain = np.random.exponential(1.0) # |H|^2 ~ Exp(1)
    power_gain = max(power_gain, 1e-10)     # prevent log(0)
    fading_dB = 10 * np.log10(power_gain)    # convert to dB
    return fading_dB
```

◆ Key insight

The fading value added to RX power can range from roughly -30 dB (deep fade — catastrophic) to +5 dB (constructive gain). In Scenario B, every node gets a different random fade, simulating the unpredictable dead zones of a rubble-filled building.

15 Received Power & SNR (Extended)

Stage 2 extends the SNR calculation to include optional Rayleigh fading and a configurable extra noise term representing jamming.

$$\text{RX Power} = \text{TX Power} - \text{Path Loss} + \text{Fading_dB}$$

$$\text{Noise Floor} = \text{noise_power_dBm} + \text{extra_noise_dB}$$

$$\text{SNR (dB)} = \text{RX Power} - \text{Noise Floor}$$

```
def calculate_snr(dist, n_exp, apply_fading=False, extra_noise_dB=0, freq_band='2.4
GHz'):
    pl      = get_indoor_path_loss(dist, n_exp, freq_band)
    rx_power = tx_power_dBm - pl
    if apply_fading:
        rx_power += get_rayleigh_fading_dB()
    actual_noise = noise_power_dBm + extra_noise_dB
    return rx_power - actual_noise, rx_power
```

Worked Examples

Scenario A at 50 m	Value
TX Power	20 dBm
Path Loss (n=3)	91 dB
RX Power	-71 dBm
Noise Floor	-77 dBm (no jamming)
SNR	~6 dB — Marginal, BPSK/QPSK only

Scenario B at 50 m	Value
TX Power	20 dBm
Path Loss (n=4)	108 dB
RX Power (pre-fade)	-88 dBm
Rayleigh Fading	e.g. -12 dB -> -100 dBm
Noise Floor (+15 dB)	-77 + 15 = -62 dBm (jammer active)
SNR	-100 - (-62) = -38 dB — TOTAL OUTAGE

◆ Why +15 dB jamming is so devastating

Adding 15 dB to the noise floor reduces SNR by exactly 15 dB on every single link simultaneously, regardless of distance. A node at 40 m with SNR = 10 dB now has SNR = -5 dB — below the minimum threshold for even BPSK (~2 dB for BER < 5%).

16 BER Formulas & Adaptive Rate Selection

Modulation	Bits/Symbol	BER Formula	SNR for BER < 5%
BPSK	1	$0.5 \times \text{erfc}(\sqrt{\text{SNR}_{\text{lin}}})$	~2 dB
QPSK	2	$0.5 \times \text{erfc}(\sqrt{\text{SNR}_{\text{lin}}})$	~2 dB
16-QAM	4	$(3/8) \times \text{erfc}(\sqrt{2 \times \text{SNR}_{\text{lin}}/5})$	~9 dB
64-QAM	6	$(7/24) \times \text{erfc}(\sqrt{\text{SNR}_{\text{lin}}/7})$	~15 dB

map_snr_to_throughput()

This function is the core of Stage 2 rate adaptation. It scans all 4 WiFi modes and returns the best achievable throughput, mode name, and BER for a given SNR. The BER acceptance threshold is 5% (more lenient than 1% in Stage 1).

```
def map_snr_to_throughput(snr_dB):
    snr_lin = 10 ** (snr_dB / 10)
    best_tp, best_mode, min_ber = 0, None, 1.0
    for std, mod, rate in wifi_modes_s2:
        p_err = ber_func[mod](snr_lin)
        tp = rate * (1 - p_err)
        if p_err < 0.05 and tp > best_tp:
            best_tp, best_mode, min_ber = tp, f'{std} {mod}', p_err
    if best_tp == 0: return 0, 'No Link', 1.0
    return best_tp, best_mode, min_ber
```

SNR Range	Selected Mode	Typical Throughput	Reason
> 17 dB	802.11n 64-QAM 54 Mbps	~51-54 Mbps	BER well below 5%
11-17 dB	802.11g 16-QAM 24 Mbps	~22-24 Mbps	64-QAM BER exceeded
4-11 dB	802.11b QPSK 2 Mbps	~1.5-2 Mbps	16-QAM too noisy
< 4 dB	No Link	0 Mbps	No modulation survives

17 Scenario A — Normal Indoor Environment

Scenario A models a standard indoor WiFi deployment — a typical office, home, or campus building. It represents the best expected real-world performance.

Parameter	Value	Meaning
Path Loss Exp. n	3	Standard office/home — walls reduce signal normally

Extra Noise	0 dB	No interference — noise floor stays at -77 dBm
Rayleigh Fading	OFF	No multipath fading — channel is deterministic
Effective Noise	-77 dBm	Baseline thermal noise only

Expected Results by Distance

Distance	SNR Range	Mode Selected	Throughput
0-20 m	20-30 dB	802.11n 64-QAM	~50-54 Mbps
20-30 m	12-20 dB	16-QAM or 64-QAM	~24-50 Mbps
30-40 m	10-18 dB	16-QAM or QPSK	2-24 Mbps
~50 m	6-10 dB	QPSK / BPSK	1-2 Mbps
> 50 m	< 4 dB	No Link	0 Mbps

18 Scenario B — War-like / Worst-Case

Scenario B simulates an extreme environment — a combat zone, disaster site, or bombed building with active radio jamming, dense concrete and rubble obstruction, and severe multipath reflections. All three impairments are active simultaneously.

Parameter	Value	Meaning
Path Loss Exp. n	4	Dense rubble, thick concrete — rapid signal absorption
Jamming	+15 dB	Active jammer raises noise floor from -77 to -62 dBm
Rayleigh Fading	ON	Severe multipath — each node gets a different random fade
Effective Noise	-62 dBm	Jammer active — RX power must exceed -62 dBm for SNR > 0

Combined Effect at 50 m

Stage	SNR Value	Change
Normal SNR (Scenario A)	~6 dB	—
After $n=4$ path loss	~-11 dB	-17 dB
After +15 dB jammer	~-26 dB	-15 dB
After Rayleigh (-10 dB)	~-36 dB	-10 dB
Final Assessment	-36 dB	TOTAL OUTAGE — No Link

◆ Anti-Jamming Frequency Hop

When SNR < 5 dB under jamming, the AP fails to receive the 3-bit STA identification code and automatically switches to the 5 GHz band (jammer-free). Communication is restored with a 1-second switchover penalty, reducing effective throughput by 10% over a 10-second window.

19 Comparative Analysis & Plots

After both scenarios are computed for all 14 STAs, results are sorted by distance and visualised in two side-by-side charts.

Chart	X-Axis	Y-Axis	What to Look For
Throughput vs Distance	Distance from AP (m)	Achievable Throughput (Mbps)	Green (A) drops gradually. Red (B) drops sharply.
SNR vs Distance	Distance from AP (m)	Received SNR (dB)	Both decrease with distance. Scenario B is 25-40

How to Read the Throughput Chart

- **Green line at 50+ Mbps:** Node is using 64-QAM — excellent signal quality
- **Step down to ~24 Mbps:** 64-QAM BER exceeded threshold — device dropped to 16-QAM
- **Step down to ~2 Mbps:** 16-QAM also failed — device is on QPSK
- **Drop to 0 Mbps:** No modulation survived — node fully disconnected
- **Red line drops early:** Scenario B: jammer + higher path loss + fading eliminate links sooner

20 Network Survivability Report

After both scenarios complete, a quantitative comparison is printed. This is the key engineering output — it answers how resilient the network is under extreme conditions.

```
=====
NETWORK SURVIVABILITY REPORT : NORMAL vs WORST-CASE
=====
Total Stations      : 14
Max Range Tested    : 50 metres
-----
[NORMAL]    Connected Nodes : X / 14
[NORMAL]    Avg Throughput   : Y Mbps
-----
[WORST-CASE] Connected Nodes : Z / 14
[WORST-CASE] Avg Throughput   : W Mbps
CONCLUSION: Severe conditions caused P% drop in connectivity.
Total throughput capacity reduced by Q%.
```

How Metrics Are Computed

Metric	Formula	Code
Connected nodes (normal)	Count STAs where tp > 0	sum(1 for tp in tp_n if tp > 0)

Connected nodes (worst)	Count STAs where $tp > 0$	<code>sum(1 for tp in tp_w if tp > 0)</code>
Avg throughput (normal)	Mean of all normal tp	<code>np.mean(tp_n)</code>
Avg throughput (worst)	Mean of all worst-case tp	<code>np.mean(tp_w)</code>
Connectivity drop %	$(1 - \text{worst}/\text{normal}) \times 100$	Printed if worst < normal
Throughput reduction %	$(\text{normal} - \text{worst})/\text{normal} \times 100$	Printed if worst < normal

21 Key Findings — Stage 2

◆ Higher data rate = higher SNR requirement

64-QAM (54 Mbps) requires ~17 dB SNR to keep BER below 5%. At 6 dB SNR (typical for a 50 m node) its BER is effectively 1 — every bit is wrong. BPSK survives down to ~2 dB.

◆ Indoor path loss far exceeds free-space

At 50 m indoors ($n=3$), path loss = 91 dB vs free-space 74 dB — 50x more power lost. Under war-like conditions ($n=4$) the extra loss reaches 34 dB — over 2500x more power lost.

◆ Rayleigh fading creates unpredictable dead zones

A node at 30 m may randomly outperform a node at 20 m if the closer node lands in a deep fade. This statistical unpredictability cannot be captured by path loss alone.

◆ The jammer is the most uniformly destructive factor

+15 dB noise uniformly removes 15 dB of SNR from every node simultaneously. A node at 5 m drops from 30 dB to 15 dB — still connected. A node at 40 m drops from 10 dB to -5 dB — immediate outage.

◆ Rate adaptation prevents total network failure

When SNR degrades, WiFi drops to a slower, more robust modulation (BPSK at 1 Mbps) and keeps the link alive. Only when SNR falls below ~2 dB does the node disconnect entirely.

◆ Near nodes survive; far nodes are most vulnerable

Nodes within 10-15 m of the AP have enough SNR margin to absorb most impairments. Nodes at 40-50 m have little headroom — a single deep fade or the jammer alone is enough.

22 Glossary of All Terms

Access Point (AP)	The WiFi router. Transmits to and receives from all Stations. Fixed at (5,5) in Stage 2.
Station (STA)	Any WiFi client device — phone, laptop, sensor. Stage 2 has 14 STAs randomly placed.
dBm	Decibels relative to 1 milliwatt. Unit for absolute power. TX = 20 dBm = 100 mW.
dB	Decibels. Unit for a power ratio. Used for path loss, SNR, and fading values.
Path Loss	Total signal attenuation between transmitter and receiver. Increases with distance and obstacles.
Path Loss Exponent (n)	Controls how fast path loss grows with distance. $n=2$ (free space), $n=3$ (office), $n=4$ (rubble).
PL_0	Reference path loss at 1 m from the AP. 40 dB at 2.4 GHz, 46 dB at 5 GHz.
Noise Floor	Baseline random electrical noise inside every receiver. Set to -77 dBm (thermal noise).
SNR	Signal-to-Noise Ratio. $SNR = RX \text{ Power} - \text{Noise Floor (in dB)}$. Higher = cleaner signal.
RX Power	Signal power arriving at the receiver. $RX = TX - \text{Path Loss} + \text{Fading (dBm)}$.
Rayleigh Fading	Random instantaneous signal variation caused by multipath reflections combining constructively or destructively.
Multipath	Signal reaching the receiver via multiple reflected paths — walls, floors, furniture, people.
Deep Fade	Destructive multipath interference causing sudden SNR drop of 20-30 dB.
Jammer	Device that transmits noise to raise the effective noise floor. Modelled as $\text{extra_noise_dB} = +15$.
AWGN	Additive White Gaussian Noise — the standard noise model. Noise samples follow a Gaussian distribution.
BER	Bit Error Rate. Probability a single bit is received incorrectly. $BER = \text{errors} / \text{total bits}$.
erfc()	Complementary Error Function. Converts SNR to BER for AWGN channels.
BPSK	Binary Phase Shift Keying. 1 bit/symbol. Most robust modulation. Min SNR ~2 dB.
QPSK	Quadrature Phase Shift Keying. 2 bits/symbol. Same BER as BPSK per bit. More efficient.

16-QAM	4 bits/symbol. Needs ~9 dB SNR for BER < 5%. Good throughput at moderate SNR.
64-QAM	6 bits/symbol. Fastest but needs ~15 dB SNR for BER < 5%. Very fragile at low SNR.
Rate Adaptation	Automatic selection of fastest modulation whose BER stays below threshold (~5%).
Throughput	Actual delivered data rate = Nominal Rate x (1 - BER). Always <= the rated speed.
No Link	State when all modulations exceed the BER threshold. Node fully disconnected.
Euclidean Distance	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Used to calculate AP-to-STA distances.
Survivability Report	Final output. Reports connected node count and average throughput in both scenarios.
Frequency Hopping	Anti-jamming technique: switch from 2.4 GHz to 5 GHz when jammed, with 1 s penalty.